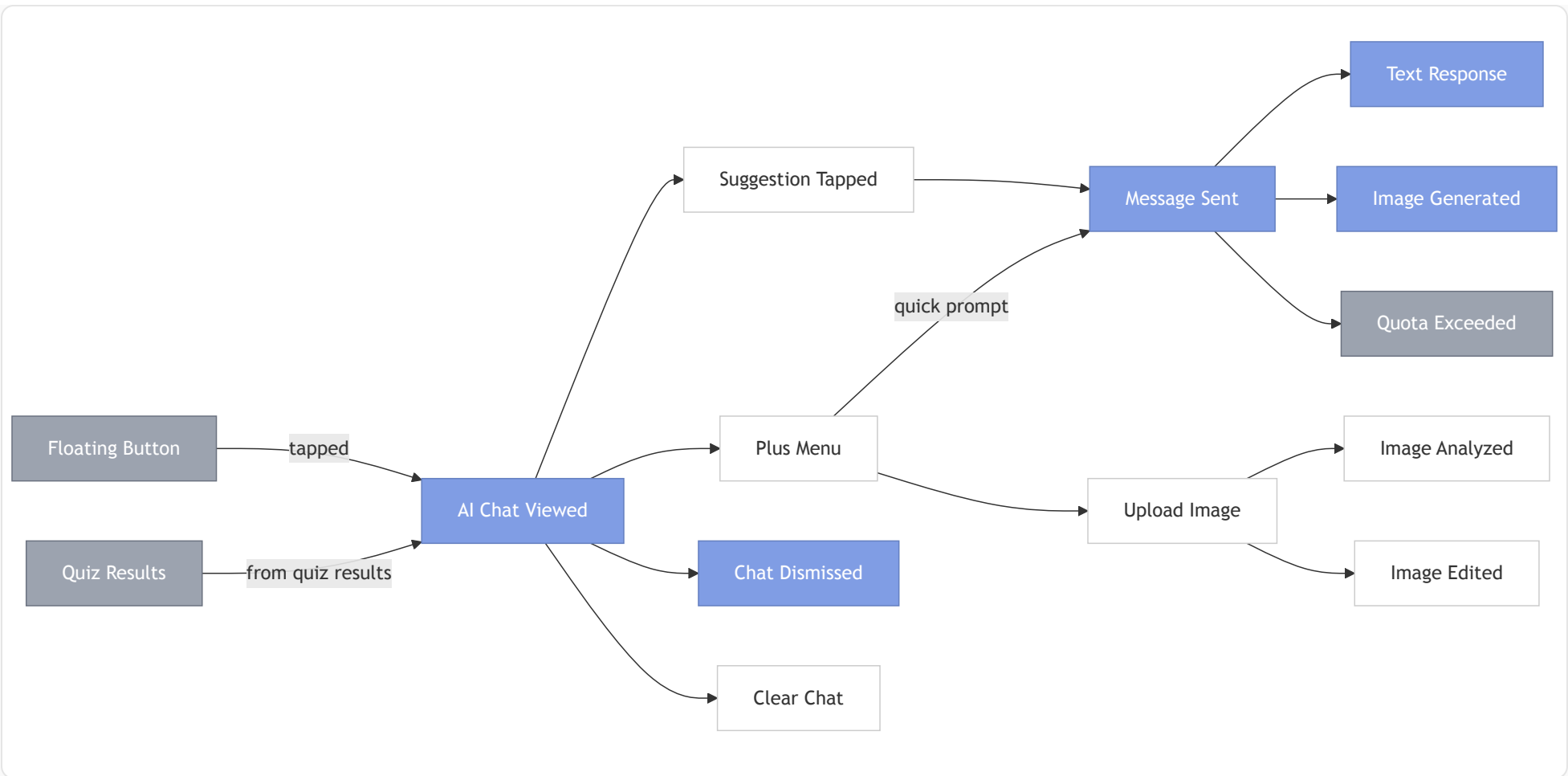


Archives — AI Chat Flow Spec

AFF-844 · PostHog Project 93650 · Updated 2026-05-06

- ☐ **Normal** — event exists in codebase
- ☐ **Red** — not in codebase, needs implementation
- ☐ **Blue** — checkpoint (key milestone)
- ☐ **Dark Grey** — entry point / alternate path

User Journey



Events

#	Node	PostHog Events	QA
1	Floating Button	ai_button_visible screen (current route pathname) ai_button_tapped screen (current route pathname)	Button not visible; button overlaps content; tap doesn't open chat; drag gesture blocks tap; button disappears after drag

#	Node	PostHog Events	QA
2	Quiz Results	quiz_results_chat_to_learn_tapped adventure_id, module_id, previous_score, era_id, era_name chat_to_learn_paywall_shown adventure_id, module_id, era_id, era_name, trigger: chat_to_learn subscribe_purchase_completed trigger: chat_to_learn subscribe_purchase_cancelled trigger: chat_to_learn chat_to_learn_response era_id, response_length	Paywall not presented; hidden context not sent to AI; quota error before API call; response not displayed; quiz context banner missing
3	AI Chat Viewed	ai_chat_viewed era_id, message_count	Modal doesn't open; analytics event not firing; era_id missing or wrong; welcome screen not shown when messages is empty
4	Suggestion Tapped	ai_suggestion_tapped suggestion_text, era_id	Suggestion buttons unresponsive; wrong suggestion text sent; suggestion tap doesn't trigger handleSend; welcome screen suggestions not shown for empty chat
5	Message Sent	ai_chat_message_sent era_id, message_type (text image), message_length, is_first_message	Send button disabled when it should be active; keyboard doesn't dismiss; empty message gets sent; input text not cleared after send
6	Text Response	ai_chat_response_received era_id, response_length, has_web_sources, web_sources_count ai_chat_error era_id ai_web_source_tapped	Response not displayed; loading spinner stuck; error message not shown; web source links broken; markdown not rendering

#	Node	PostHog Events	QA
		source_url, source_title, era_id	
7	Image Generated	ai_image_generated era_id ai_chat_error era_id	Image prompt not detected; "Generating image..." spinner stuck; generated image not displayed; base64 decode failure
8	Upload Image	ai_image_selected era_id, source: library	Permission denied on iOS; image picker doesn't launch; base64 encoding fails; pending image preview not shown; remove button doesn't clear
9	Image Analyzed	ai_image_analyzed era_id, has_question ai_chat_error era_id	"Analyzing image..." spinner stuck; analysis response empty; uploaded image not sent to API; error not surfaced
10	Image Edited	ai_image_edited era_id ai_chat_error era_id	Edit prompt not detected; "Generating image..." spinner stuck; edited image not displayed; upload to storage fails
11	Quota Exceeded	ai_quota_exceeded request_type (chat image_generate image_edit image_analyze), limit, is_subscriber	Quota error message not shown; wrong limit displayed; reset date incorrect; subscriber flag wrong
12	Plus Menu	ai_action_menu_viewed era_id ai_quick_prompt_tapped prompt_text, era_id	Menu doesn't appear; menu items unresponsive; cancel button doesn't dismiss; quick prompt sends wrong text

#	Node	PostHog Events	QA
13	Image Viewer	ai_image_shared era_id ai_image_viewer_viewed era_id, image_type (generated edited uploaded)	Image viewer blank; full-screen image not loading; share button doesn't open share sheet; temp file cleanup fails; close button unresponsive
14	Clear Chat	ai_chat_history_cleared era_id, message_count (before clear)	Menu doesn't appear; clear confirmation alert doesn't show; messages not deleted from storage; menu doesn't dismiss after action
15	Chat Dismissed	ai_chat_dismissed era_id, message_count	Modal doesn't close; messages not saved to cloud before close; back button on Android doesn't trigger close tracking

AI Quiz Explanation events (related but outside chat modal)

Event	Properties	Status
ai_quiz_explanation_requested	adventure_id, module_id, era_name, question_count	Wired in AIQuizExplanation.tsx
ai_quiz_explanation_generated	adventure_id, module_id, era_name, question_count, duration_ms	Wired in AIQuizExplanation.tsx

Questions This Flow Answers

1. Strategic Questions

- 1. How many users open the AI chat at least once per week?
- 2. How many users open AI chat from the floating button vs from Chat to Learn?

3. How many users hit the monthly quota limit?
4. How many free users hit the quota limit and then see the paywall?
5. How many users who hit the quota limit upgrade to a paid plan?
6. How many AI chat sessions result in zero messages sent (opened and dismissed immediately)?
7. How many users send image-related requests (generation, editing, analysis)?
8. How many users use the AI chat more than 5 times in a single day?

2. Behavioral Insights

1. How many users tap a suggestion on the welcome screen as their first message?
2. How many users type a custom first message instead of tapping a suggestion?
3. How many users receive a response with web search sources?
4. How many users tap on a web search source link?
5. How many users request image generation through text prompts?
6. How many users upload an image from their photo library?
7. How many uploaded images are analyzed vs edited?
8. How many users open the full-screen image viewer?
9. How many users share a generated or edited image?
10. How many users open the action menu (plus button)?
11. How many users select a quick prompt from the action menu?
12. How many users clear their chat history?
13. How many users encounter an AI chat error?
14. How many users complete a Chat to Learn session (quiz context + AI response)?
15. How many users send a follow-up message after a Chat to Learn response?

3. Checkpoint Validation

1. **AI Chat Viewed** — How many users open the chat modal?
2. **Message Sent** — How many users send at least one message?
3. **Text Response** — How many messages get a successful AI response?
4. **Image Generated** — How many users request and receive a generated image?
5. **Chat Dismissed** — How many users close the chat after at least one exchange?

Analysis Notes

- **Chat to Learn is gated by subscription:** Non-subscribers see a paywall when tapping "Chat to Learn" on quiz results. Track conversion rate from `chat_to_learn_paywall_shown` to `subscribe_purchase_completed` with trigger `chat_to_learn`.
- **Quota enforcement is server-side:** The client `checkQuota()` always returns `allowed: true`. The backend API returns HTTP 429 when the limit is hit. The `ai_quota_exceeded` event fires on the client when the backend rejects a request.
- **Image routing is heuristic:** `isImageRequest()` and `isImageEditRequest()` use keyword matching. Some prompts may be misclassified (image intent routed to text chat or vice versa).
- **Suggestion tap vs typed message:** Both go through the same `handleSend()` path and fire `ai_chat_message_sent`. To distinguish them, add `ai_suggestion_tapped` on the welcome screen.
- **Chat to Learn auto-context:** When the user taps "Chat to Learn" after a quiz, a hidden context message is sent to the AI automatically. This is not a user action. The AI response appears with a quiz context banner. The `chat_to_learn_response` event tracks this.
- **Floating button is hidden on certain screens:** Hidden on onboarding, auth, today tab, and subscribe screens. Track `ai_button_visible` to understand which screens drive the most chat opens.